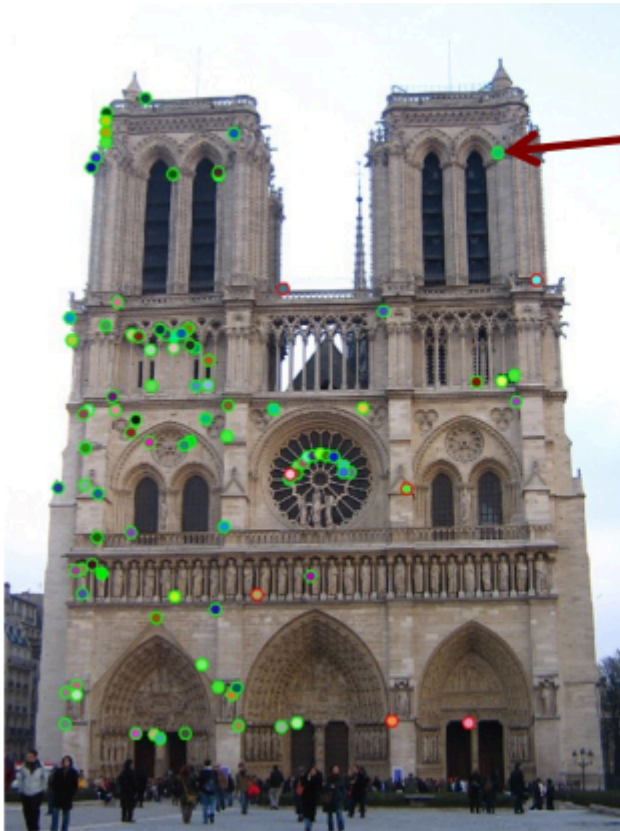


L11: Visual Features: Keypoints

- **Keypoint**: locally distinct location in an image
- Feature **descriptor**:
 - **summarises local structure around** keypoint



keypoint

descriptor at
the keypoint

$$f = \begin{bmatrix} 0.02 \\ 0.04 \\ 0.1 \\ 0.03 \\ 0 \\ \dots \end{bmatrix}$$

- Most keypoints use image gradients
- Keypoints: Finding distinct points
 1. Corners via structure matrix
 - Harris corners
 - Shi-Tomasi corner detector
 - Förstner operator
 2. Difference of Gaussians
 - Iterates over scales and blur
 - Finds corners and blobs

Corners

- Often locally distinct points
- Invariant to translation, rotation, illumination
- **Corner**: 2 edges in roughly orthogonal directions

- **Edge:** sudden brightness change

Finding Corners

- Search for intensity changes in 2 directions
- Compute SSD of neighbor pixels around (x, y)

$$f(x, y) = \sum_{(u,v) \in W_{xy}} (I(u, v) - I(u + \delta u, v + \delta v))^2$$

**local patch
around (x, y)**

**sum of squared differences
of image intensity values of
pixels under a given shift
($\delta u, \delta v$)**

- Using Taylor expansion, we obtain

$$I(u + \delta u, v + \delta v) \approx I(u, v) + [J_x \quad J_y] \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$

- Integrated and in Matrix form

$$f(x, y) \approx \sum_{(u,v) \in W_{xy}} [\delta u \quad \delta v] \begin{bmatrix} J_x^2 & J_x J_y \\ J_x J_y & J_y^2 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$

$$f(x, y) \approx [\delta u \quad \delta v] \begin{bmatrix} \sum_W J_x^2 & \sum_W J_x J_y \\ \sum_W J_x J_y & \sum_W J_y^2 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix}$$

$$M = \begin{bmatrix} \sum_W J_x^2 & \sum_W J_x J_y \\ \sum_W J_x J_y & \sum_W J_y^2 \end{bmatrix}$$

- Jacobians computed via a convolution with a gradient kernel such as Scharr or Sobel operator:

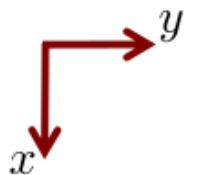
$$J_x^2 = (D_x * I)^2$$

$$J_x J_y = (D_x * I)(D_y * I)$$

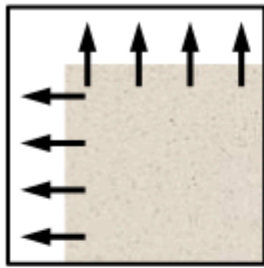
$$J_y^2 = (D_y * I)^2$$

$$D_x^{\text{Scharr}} = \frac{1}{32} \begin{bmatrix} 3 & 10 & 3 \\ 0 & 0 & 0 \\ -3 & -10 & -3 \end{bmatrix} \quad D_x^{\text{Sobel}} = \frac{1}{8} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

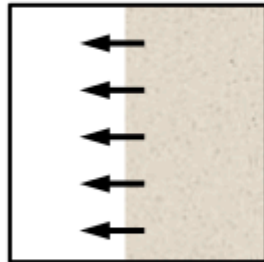
$$D_y^{\text{Scharr}} = \frac{1}{32} \begin{bmatrix} 3 & 0 & -3 \\ 10 & 0 & -10 \\ 3 & 0 & -3 \end{bmatrix} \quad D_y^{\text{Sobel}} = \frac{1}{8} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$



- Summarise the dominant directions of the gradient around a point



$$\Rightarrow M = \begin{bmatrix} \gg 1 & \sim 0 \\ \sim 0 & \gg 1 \end{bmatrix} \quad \text{YES!}$$



$$\Rightarrow M = \begin{bmatrix} \sim 0 & \sim 0 \\ \sim 0 & \gg 1 \end{bmatrix} \quad \text{NO!}$$



$$\Rightarrow M = \begin{bmatrix} \sim 0 & \sim 0 \\ \sim 0 & \sim 0 \end{bmatrix} \quad \text{NO!}$$

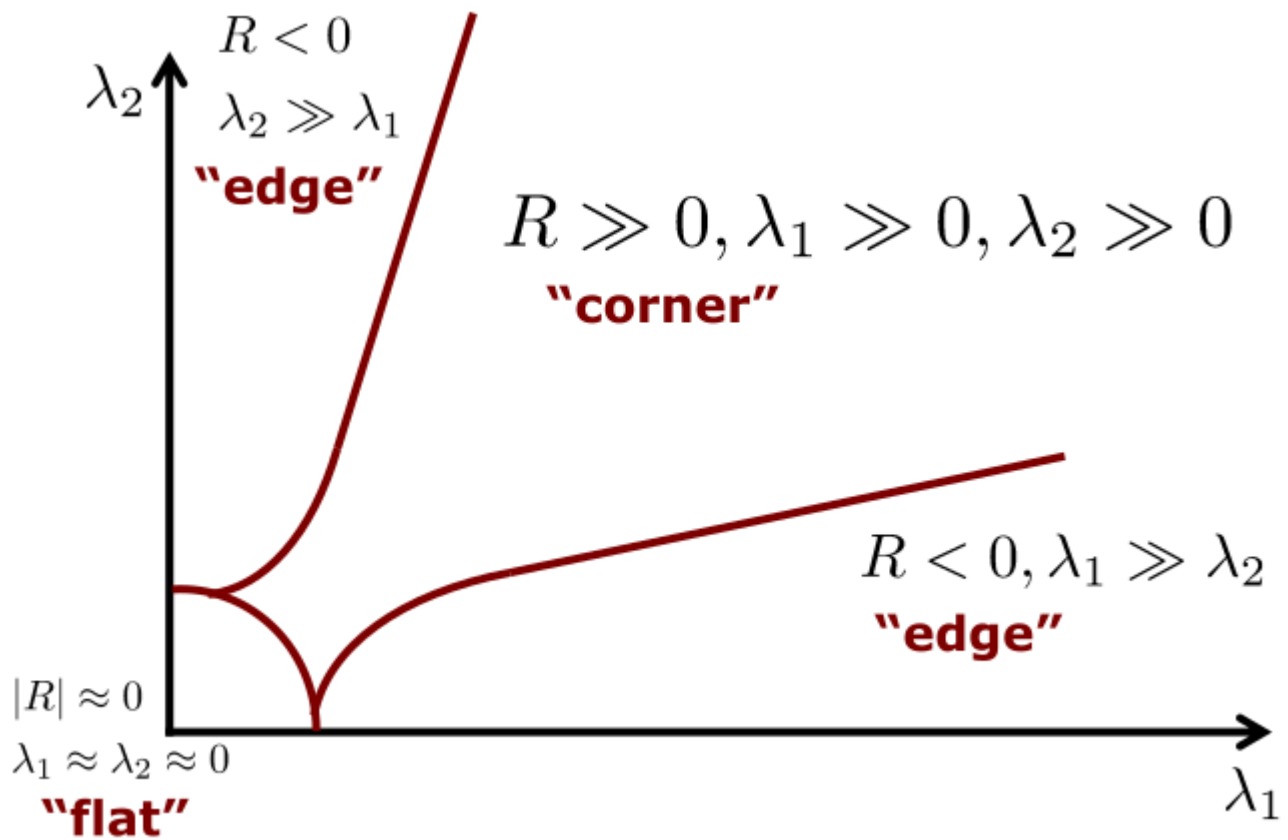
- Key idea: Consider a point as corners if its structure matrix has 2 large Eigenvalues
- Approaches rely on structure matrix
- Use different criteria for deciding if point is a corner
- Förstner offers subpixel estimation

Harris Corner Criterion

- Criterion

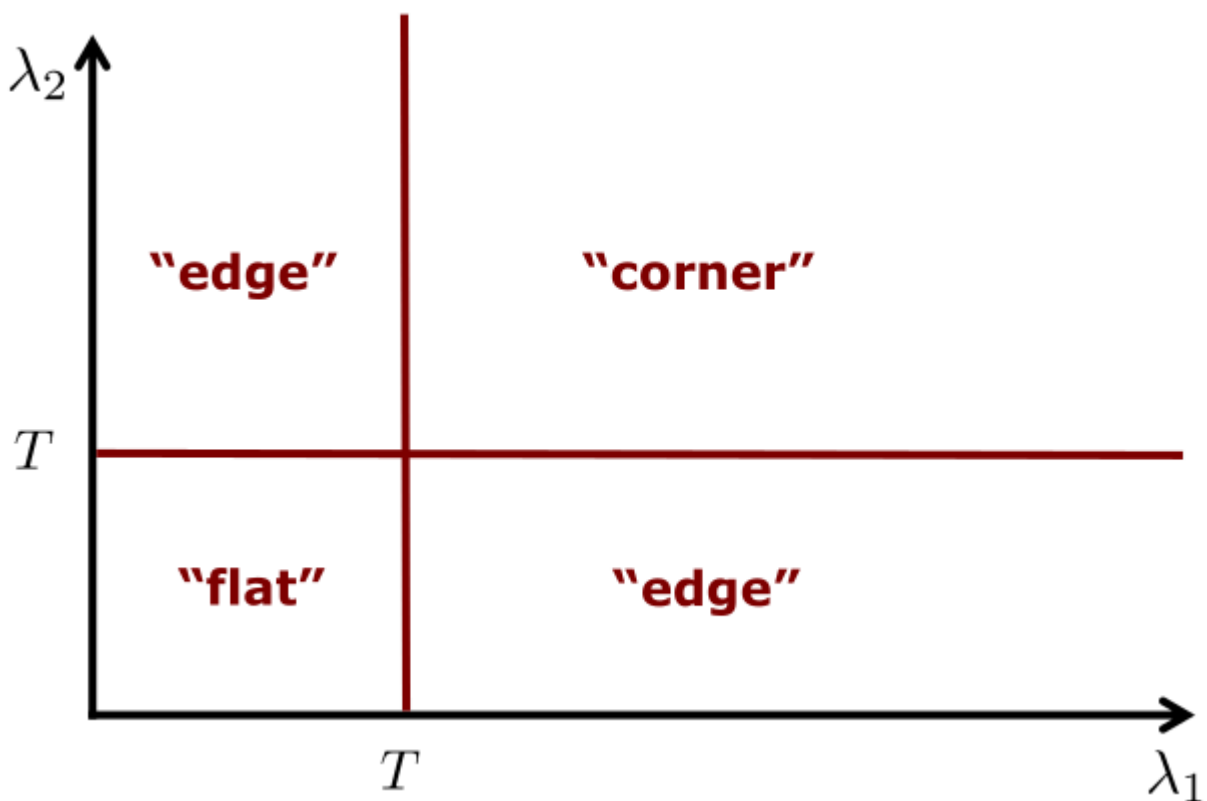
$$R = \det(M) - k(\text{trace}(M))^2$$

- $= \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$
 - with $k \in [0.04, 0.06]$



Shi-Tomasi Corner Detector

- Criterion: Threshold smallest Eigenvalue
- $\lambda_{\min}(M) = \frac{\text{trace}(M)}{2} - \frac{1}{2} \sqrt{(\text{trace}(M))^2 - 4\det(M)}$
 - $\lambda_{\min}(M) \geq T$: corner

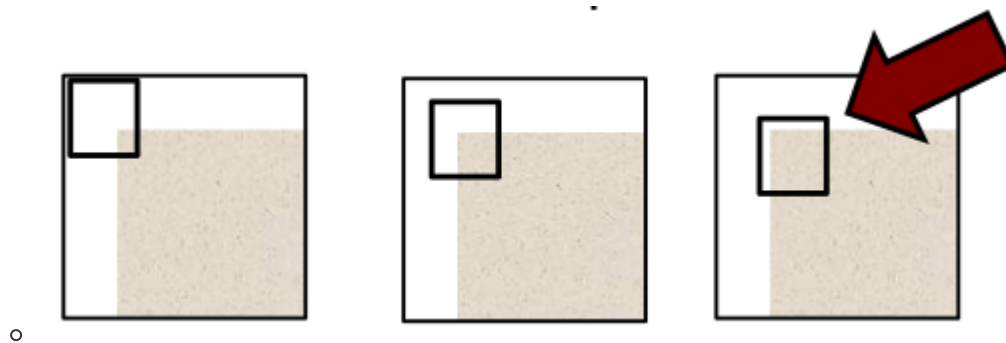


Förstner Operator Criterion

- Similar to harris corner detector
- Defined on the inverse of the M (covariance matrix of possible shifts)
- Similar criterion on size and roundness of the error ellipse of covariance matrix
- Extension for sub-pixel estimation

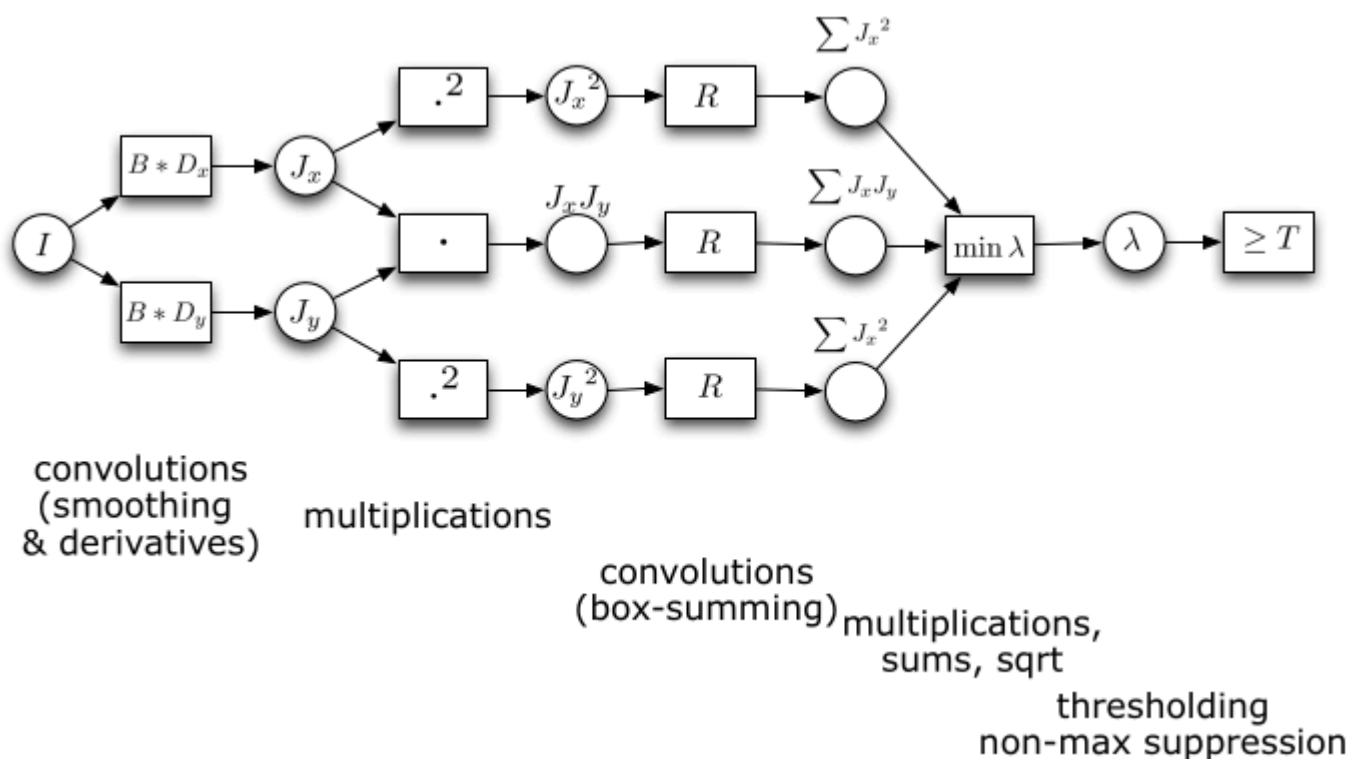
Non-Maxima Suppression

- Within local region:
 - position with the maximum value R or λ_{min} and select



- RGB to gray-scale conversion first
- Real images are affected by noise
 - smoothing of the input is suggested

Summary Corner Detection



- All 3 detector perform similarly

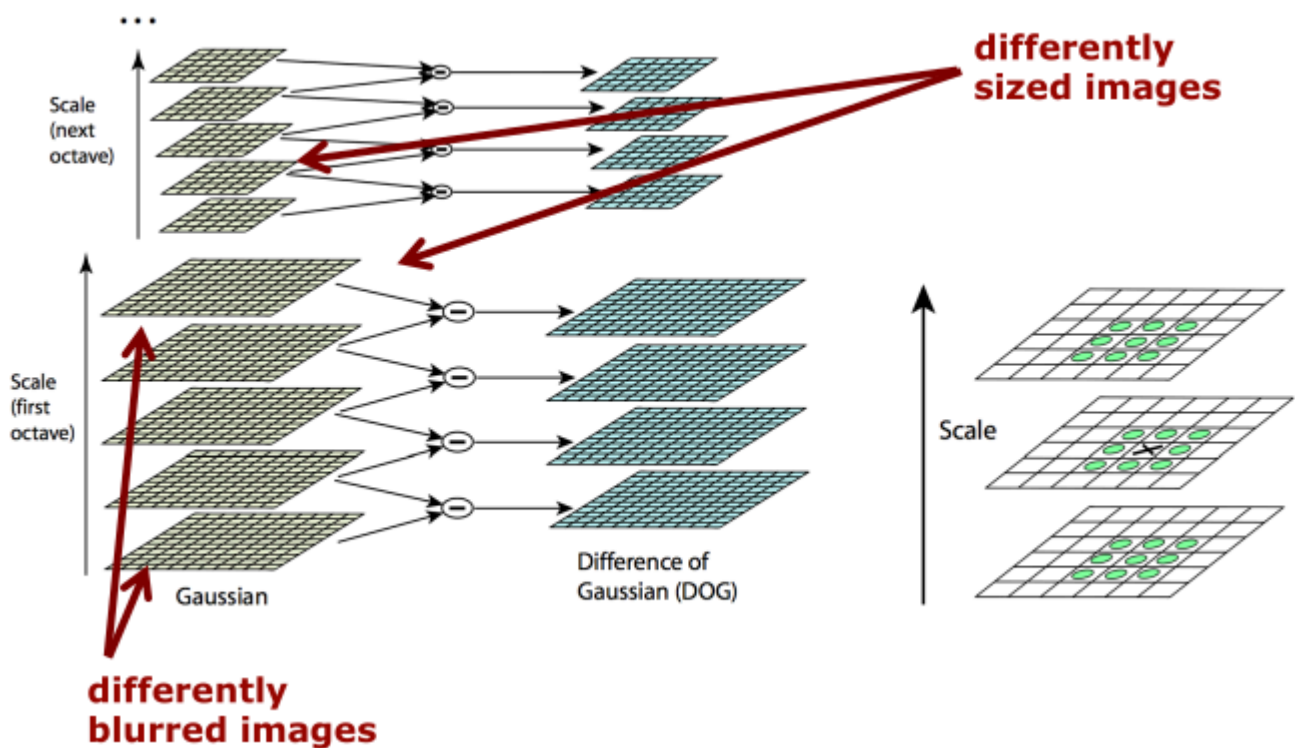
- Förstner first and described subpixel estimate
- Shi-Tomasi slightly outperforms Harris corners
 - -> default corner detector

Difference of Gaussians

- Variant of corner detection
- Responses at corners, edges, and blobs
 - Blob: mainly constant region but different to its surroundings

Procedure

- Over different image pyramid levels
1. Gaussian smoothing
 2. Difference-of-Gaussians: find extreme (over smoothing scales)
 3. Maxima Suppression at edges



Difference of gaussians

- Subtract differently blurred images from each other



-
- Increase visibility of corners, edges, other details present in the image



-
- Blurring: Filters out High-frequencies (noise)
- Subtracting differently blurred images from each other only keeps the frequencies that lie between the blur level of both
- DoG acts as a band-pass filter

Extrema Suppression

- DoG finds blob-like and corner-like image structures but also:
 - strong responses along edges
 - Edges are bad for matching
 - Eliminate edges via Eigenvalue test (similar to Harris corners)